Faizah Binte Naquib

Applied Machine Learning
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Applied ML scientist who enjoys turning messy, real-world problems into working solutions. My experience spans OCR research, autonomous-vehicle perception and edge deployment, with a focus on understanding failures and testing practical improvements.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL
Software: React, REST APIs, Git, Linux, Docker, unit and integration testing
Applied ML: PyTorch, OpenCV, OCR, ROS 2, Autoware, model evaluation, ONNX/RKNN

EXPERIENCE

Correct-AI | Machine Learning Scientist — Aug 2025 - Present

- Develop ROS 2/Autoware perception and simulation workflows with camera-LiDAR inputs and Unity/AWSIM digital twins; integrate sensors, vehicle models and recorded data for autonomous-system testing.
- Build repeatable perception evaluations across KITTI, Waymo and T4 datasets; compare LiDAR-only and camera-LiDAR fusion, and diagnose timing, coordinate-frame and ground-truth mismatches.
- Train and optimize detection models for edge inference using ONNX/RKNN conversion and performance profiling.
- Develop terrain-aware traversability mapping and vehicle-feasibility checks to support autonomous navigation in off-road environments.

Synapsis Medical Technologies | ML Research Assistant — May - Aug 2025

- Designed OCR and document-understanding pipelines for noisy clinical documents, including parsing and structured extraction; worked on retrieval-augmented generation and prompt engineering.

Amii | WILO Participant — May - Aug 2025

- Translated startup needs into ML problem definitions and prototype plans; researched a materials recommender using text embeddings and vector search, and prepared evaluation recommendations.

Optimizely | Software Engineer Intern — Mar - Aug 2022

- Implemented authentication workflows, contributed to an AngularJS-to-React migration, and collaborated with QA on testing and debugging.

Thrive EdTech | Tech Intern — Jul - Oct 2021

- Built React and REST API features and prototyped OCR-based digitization of textbook content.

RESEARCH

University of Alberta | MSc Thesis - OCR Optimization — 2023 - Jan 2025

- Trained a transformer-based stand-in for a black-box OCR engine, enabling an image-enhancement model to learn how to improve text recognition without access to the OCR engine's internals.
- Extended the approach to whole, variable-resolution documents, eliminating word-box annotations and cutting OCR calls per training-data pass by **95.1%-98.1%** compared with word-level processing.
- Reduced Tesseract document word error rate by **58.5% on POS** and **23.5% on SROIE**, two receipt-image datasets, compared with reading the original, unprocessed images.

SELECTED PROJECT

Interactive OCR and Text Refinement Demo | JavaScript, Python, AWS Lambda

- Built a canvas-input OCR demo with PaddleOCR-to-Tesseract.js fallback, processing-status UI, and a Python AWS Lambda endpoint for Llama 3.1 text refinement.

EDUCATION

MSc, Computing Science - University of Alberta | 2025
BSc, Computer Science and Engineering - Ahsanullah University of Science and Technology | Aug 2021
Publications: Finger-vein biometrics - ICCIT 2020 | ICAICT 2020